Others: (6 Marks) Section 1: OS Fundamentals and Concurrent Execution. Each Question 1/2 Mark

Classify each statement as true or false

1) System calls do not change the privilege mode of the processor.	~~a)~~ True ~~b)~~ False
2) The windows operating system is not responsible for resource allocation between competing processes.	~~a)~~ True ~~b)~~ False
3) A smaller page size leads to smaller page tables.	a) True b) False
4) A smaller page size leads to more TLB misses.	a) True b) False
5) Semaphore has 2 non-interruptible operations; wait () and signal ().	a) True b) False
6) Priority Inversion Problem occurs if a low priority job, though ready, is never scheduled to run because there is always a higher priority job ready.	a) True b) False
7) The busy waiting can waste CPU time.	a) True b) False
8) A smaller page size leads to fewer page faults.	a) True b) False
9) Blocking implementation of the semaphore mechanism needs waiting queue	a) True b) False
10) In Peterson's solution, release and acquire does not need to be atomic operations	a) True b) False
11) Counting semaphores are used for solving mutual exclusion problem.	a) True b) False
12) A deadlocked state occurs whenever a process is waiting for I/O to a device that does not exist.	a) True b) False

CLO 6/SLO I: (4 Marks) Section 2: Memory Management Each Question 1/2 Mark

13) The page table contains the frame num for each page of a process.	a) base address b) limit address c) frame numbers d) None of the above
14) In contiguous memory allocation with fixed sized partition, the degree of multi programming is bounded by ___________ NOP	a) the number of partitions b) the CPU utilization c) the memory size d) all of the above
15) Page faults occur when _________________	a) a process references a page that exist in main memory

$2^{16} = 65536/1028 = 64KB$	b) a process references a segment that is not in main memory c) a process references a page that that does not have a frame number stored in page table d) All of the Above
16) In a 16-bit machine with page size 1K byte and configured with maximum adequate frames. Which program size is impossible to load into memory managed by paging technique?	a) 16 KB < 64 KB b) 5.5 KB < 64 KB c) 32000 Bytes < 64 KB d) 68 KB > KB
17) What is compaction?	a) a technique for overcoming internal fragmentation b) a paging technique c) a technique for overcoming external fragmentation d) a technique for overcoming fatal error
18) In a 16-bit machine with 4k page size. What is the maximum number of pages that can be addressed in this system? $2^{16} = 64\ KB / 4\ Kbyte = 16$??? 8	a) 16 b) 4 c) ~~8~~ d) 32
19) If a page number is not found in the TLB, then it is known as a :	a) TLB miss b) Buffer miss c) TLB hit d) All of the mentioned
20) In Paging Technique every address generated by the CPU is divided into two parts :	a) frame bit & page number b) page number & page offset c) page offset & frame bit d) frame offset & page offset

CLO 8/SLO I: (Marks 3) Section 3: Disk Fundamentals

Each Question has 1/2 Mark

21) A logical block can be smaller than the sector size of a disk.	F
22) Disk Linked allocation method ensures that only one access is needed to get a disk block using direct access.	F
23) File whose blocks were allocated using Indexed allocation method needs 2 disk block access in order to read or write any byte of the file.	T
24) On 32-bit machine a FAT file system can't be used to format a terabyte hard disk as one partition.	T

25) Which of the following cannot be considered as disk access delay?	a) transfer delay b) rotational delay c) seek delay d) cache memory access delay.
26) Disk file system with contiguous allocation is suitable for formatting …………..?	a) CDROM b) Video tape c) Hard Disk d) Flash memory sticks

Others: (6 Marks) Section 4: Process Scheduling

(Each question 1.5 Mark)

27) **What is an Address binding means?**

Address binding means converting the (relative or symbolic) address used in a program to an actual physical (absolute) RAM address. It can happen at three different stages; compile time, load time and run time.

28) **Mention 2 process states and explain how to enter and leave such 2 states.**

- **new: The process is being created (may be on disk in special area). Enter once created and left if moved to memory.**
- **running: registers loaded, Instructions are being executed. Enter if scheduled and left if quantam finished or on i/o instruction or terminated.**
- **waiting: The process is waiting for some event to occur (I/O). enter if instruction executed is I/O and left on device interrupt.**
- **ready: runnable, but waiting to be assigned to a processor. Enter if new process put into memory and left if schedule choose it to run**
- **terminated: The process has finished execution. Enter if process successfully terminated of on fatal error.**

29) **What would happen if round robin quantum was chosen too long, and what would happen if it was too short?**

- Setting quantum too short → many process switches → lowers the CPU efficiency.

- Setting the quantum too long → may cause poor response time to short interactive requests

30) **Mention four CPU scheduling events when scheduling can occur during the life cycle of a process?**

- **CPU scheduling occurs when**
  - **1) A process is created and is put in a ready state**
  - **2) A process voluntarily *yields the CPU***
  - **3) A process switches from running to waiting state e.g. blocking on I/O**
  - **4) A process switches from running to ready state e.g. because it is interrupted and its timeslice expires**
  - **5) A process switches from waiting to ready state e.g. an interrupt signifies that I/O is complete.**
  - **6) A process terminates**

CLO 6/ SLO I: (Marks 5) Section 5: Memory Management
Each has 1 Marks

13) Describe the difference between external and internal fragmentation. Indicate which of the two are most likely to be an issue on a) a simple memory management machine using base limit registers and static partitioning, and b) a similar machine using dynamic partitioning.

External Fragmentation – total memory space exists to satisfy a request, but it is not contiguous. All the holes are too small to fit the requirement.
Internal Fragmentation – allocated memory may be slightly larger than requested memory; this size difference is memory internal to a partition, but not being used

Fixed size partitions suffers from internal fragmentations and decreased number of multiprogramming. But Variable size partitions increases degree of multiprogramming but suffers from external fragmentations. Both techniques uses memory management unit with base limit registers.

14) Given the reference string of page accesses:
1 2 3 4 2 3 4 1 2 1 1 and a system with three page frames, what is the final configuration of the three frames after the true LRU algorithm is applied?

1 1	1	1 1	~~4~~ 4	4	4	4	4	4 4	4	4	
	2 2	2 2	2 2	2	2	2	~~1~~ 1	1 1	1	1	
		3 3	3 3	3	3	3	~~3~~	2 2	2	2	

F F	F F	F F	F				F	F			

15) Make a comparison between Paging and Segmentation memory techniques and show the advantages and disadvantages?

Paging does not suffer from external fragmentation but suffer from internal fragmentation in the last page. It has fast allocation mechanism because programs are divided into equally size pages.
Segmentation are natural because programs are divided into Segments, which are constructed according to user semantics.
Segmenation are good in sharing and therefore it can utilize the memory resorce effictifly.

Both paging and segmentation is suffering from the slow memory access and storage to save table (page table and segment table).

0.98 + 0.02
× ×
Hit= 20+100=120 | Miss= 20+100+100 = 220

Later **16) In a paging system assume it takes 20 nsec to search TLB and 100 nsec to access memory. 98% of the time we are lucky and get the pages numbers in the TLB. Calculate the effective memory access time.**

Effective memory access time = P(hit) * hit-time + P(miss) * miss-time
= 0.98 * 120 + 0.02 * (220)

16 add 1K = 1024

17) A pure paging system uses 16-bit address and 1K pages. The following shows the page tables of Process 1. Translate the logical address 2048 of Process Fill your answers into the table below.
[1 Marks]

Process 1

0	2
4	3
5	8
2	0

8

Process	Address	Page number	Offset	Physical Address
Process 1	2048	5	0	8*1024

CLO 8/ SLO I: (Marks 6) Section 6: Disk Management

18) Mention 2 disadvantages of the linked method of disk space allocation?
[1Marks]

Random access is slow!. And Reliability is very low. If one block pointer is corrupted, we loose file after that point

19) Hard disk block size is an important issue in file system design. What are the advantages and disadvantages of choosing large or small block size [1 Marks]

-Larger block sizes => higher internal fragmentation.
- Not good for small files because they waste Disk space

-Smaller block sizes => Files span multiple blocks and this need multiple seeks and rotations to read data => reduce performance.

Thus if the allocation unit of block is too large we waste space and if it is small we waste time

BS = 1024 bytes 1046 1046 − 1024 = 22 0 − 22 23rd byte

20) Assume that the block size is 1024 bytes, fill in the following spaces [2 Marks]

1. The physical Address of byte number 1046 in a file stored using contiguous allocation is _in the 23th byte of the block number _20 Keep in mind that this file is stored in several block starting from block number 19 and end at block 24.

+1 23 byte block num 20

19 + 1046/1024
19 + 1.02 = 20

Extra

MMU : 1. Protection
2. from logical → physical address

1045-1024 +1 =22

2. The physical Address of byte number 1045 in a file stored using the following FAT is byte number 22th of block 7___. Keep in mind that the starting entry for this file is block 4.

22nd byte 1045-1024 =21+1 22nd byte

		8		7			2	-1	
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

block 7

4+1

21) For A Linux file system with i-node of 13 pointers and uses a block of size 1 Kbytes, consider a file of size 11.5 Kbytes. For such file, fill in the space in the following:

[2 Marks]

i. How many sectors are allocated for such file if sector size is 1024 bytes? 12
ii. How many disk access do you need to read byte number 4097? 1